

CariSurg MedTech Pathways Research Project

New Preliminary Proposal

Project Title: AI-Assisted Triage Support for Earlier Identification of High-Risk Patients in a Caribbean Emergency Department: A Machine Learning Pilot

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Link to github ED workflow diagram: [ED workflow diagram](#)

Statement of the Problem:

Triage nurses make high-stakes decisions in fast-moving emergency departments, where a missed or delayed assessment can directly affect patient outcomes [1,2]. In Caribbean hospitals, this challenge is further complicated by high patient volumes, limited inpatient beds, and inconsistent access to digital patient records, making it harder to consistently identify who needs urgent care first [1]. Machine learning tools have been developed to support triage decisions in well-resourced hospitals in North America, Europe, and other non-Caribbean settings, with some showing strong results in predicting acuity, admission risk, ICU need, and patient disposition [3–11]. However, these tools have not been tested in Caribbean emergency departments, where workflows, data availability, and resource constraints are different [1,3,4]. This project tests whether an AI tool can help triage nurses flag high-risk patients earlier, without replacing their clinical judgment or disrupting the existing workflow.

Previous Work

Previous research shows that AI-assisted triage has strong potential to support triage decision-making, but important gaps remain for Caribbean implementation. The following studies are summarised using four key points: the problem addressed, the method used, the main outcome, and the limitation identified.

De Freitas et al. (2020) Emergency departments in the Caribbean face patient flow problems that go beyond clinical decision-making. De Freitas et al. conducted 203 hours of direct observation and process mapping at a tertiary hospital in Trinidad and Tobago to document how patients moved through the department. They found that nursing shortages, limited inpatient beds, scarce supplies, and poor coordination between departments were the main drivers of delays. The authors noted that the study was descriptive and did not test any intervention, leaving open the question of whether tools such as AI-assisted triage could improve flow in this setting [1].

Tyler et al. (2024) Traditional five-level triage systems rely heavily on nurse judgment, which can be inconsistent under pressure. Tyler et al. conducted a scoping review using Joanna Briggs Institute

methodology, searching three major databases for US-based peer-reviewed studies on AI in emergency department triage published between 2013 and 2023, screening 1,140 citations down to 29 included studies. They found that machine learning models consistently outperformed conventional triage systems in predictive accuracy, disease identification, risk assessment, and resource allocation. However, the review was limited to studies conducted in the United States, which reduces how well the findings apply to healthcare systems with different staffing structures, patient populations, and resource availability, including Caribbean emergency departments [2].

Da'Costa et al. (2025) Emergency departments need triage tools that are consistent and responsive in real time, but it is unclear whether AI can reliably deliver this across different settings. Da'Costa et al. reviewed published literature on AI-driven triage systems in emergency departments, assessing their reported benefits and barriers across multiple healthcare contexts. They found that AI-driven tools showed promise in improving triage consistency and supporting real-time decision-making, but also identified significant barriers including data quality problems, algorithmic bias, regulatory challenges, and difficulty integrating these tools into existing clinical workflows. The authors noted that most reviewed systems lacked rigorous clinical validation, making it unclear how safely they could be adopted outside the settings where they were developed [3].

Tahernejad et al. (2024) AI has been proposed as a tool to support triage decisions in both routine emergency care and mass casualty events, but it is not clear how consistently it performs across different environments. Tahernejad et al. conducted a systematic review searching multiple databases for studies on AI applications in emergency and disaster triage. They found that AI tools showed promise across a range of high-pressure triage settings, but stressed that careful local validation is essential before any tool is implemented in a new environment. The authors highlighted that most included studies came from high-income settings and that performance in lower-resource or differently structured health systems, such as those in the Caribbean, had not been established [4].

Agius et al. (2025) Triage nurses must make several decisions at once, including how urgent a patient is, whether they will need admission, and where in the hospital they should go. Agius et al. trained and tested machine learning models on emergency department data to predict Emergency Severity Index level, hospital admission, and admitting ward simultaneously. They found that machine learning could accurately support all three decisions using data available during the ED visit, showing that a single model can serve both clinical and operational planning purposes. The authors noted that the model was developed and validated at a single centre, which limits how confidently the findings can be applied to other emergency departments with different patient populations or workflows [5].

Chen et al. (2023) Identifying patients who will need intensive care early in their ED visit is a high-stakes problem that current triage systems do not always solve reliably. Chen et al. conducted a systematic review of machine learning models developed to predict ICU admission from emergency department data, assessing model types, input variables, and performance across multiple studies. They found that logistic regression, random forest, and gradient boosting approaches all achieved strong performance in identifying critically ill patients, suggesting that multiple model types are viable for this task. However, the authors noted that most included models were developed in single-centre studies with complete electronic records, raising questions about how well they would perform in settings with incomplete or paper-based data [6].

Pandey et al. (2025) Patients who need intensive care are not always identified quickly enough in the emergency department, which can delay treatment and worsen outcomes. Pandey et al. developed and tested a machine learning model to predict early ICU admission among ED patients, focusing on variables available at or shortly after triage including triage category, vital signs, and ambulance arrival status. They found that these early variables were among the strongest predictors of ICU need, showing that useful prediction is possible before a full clinical assessment is completed. The authors acknowledged that the model was developed at a single Australian hospital and that its performance in other settings, particularly those with different data availability or patient populations, would need to be tested separately [7].

Lee et al. (2021) It is not known how well hospital admission can be predicted using only the limited information available at triage, particularly in settings where full patient records are not accessible. Lee et al. trained and tested machine learning models on emergency department data to predict hospitalisation among urgent ED patients using only a small number of triage-available variables, without relying on laboratory results or patient history. They found that hospitalisation could be predicted with strong accuracy even with this limited input, suggesting that useful AI triage tools do not need to depend on complete data. The authors noted that the model was developed and validated at a single hospital, and that performance in other settings including those with different documentation practices had not been tested [8].

Araouchi and Adda (2024) Triage in busy health centres is inconsistent and time-pressured, and AI has been proposed as a way to support more reliable patient prioritisation. Araouchi and Adda developed TriageIntelli, an AI-assisted multimodal triage system, training seven machine learning models including support vector machines, gradient boosting, and a stacking ensemble on a publicly available triage dataset to predict patient priority level. The stacking model achieved 80% accuracy and an AUC of approximately 0.93, with the best individual models performing only slightly below this. The authors acknowledged that the study relied on a single dataset, which may limit how well the models generalise to other clinical environments with different patient demographics, seasonal patterns, and hospital contexts [9].

El Ariss et al. (2024) Emergency departments struggle to allocate limited clinical resources efficiently, particularly when patient demand is unpredictable. El Ariss et al. developed and validated a machine learning framework using ED data to predict patient resource needs beyond the initial triage decision, testing whether AI could support earlier and more accurate resource planning. They found that the framework improved resource allocation decisions compared to standard clinical processes, showing that AI can have operational value in the ED beyond individual patient risk scores. The authors noted that the model was validated at a single centre and that its performance in EDs with different resource constraints or documentation practices remained untested [10].

Chang et al. (2024) Structured triage data alone may not capture everything a nurse observes during a patient assessment, and free-text nursing notes could add useful clinical information for AI prediction. Chang et al. applied machine learning and natural language processing to emergency department records, combining structured triage variables with unstructured nursing note text to predict clinical disposition. They found that adding free-text nursing notes to structured triage data improved prediction accuracy for clinical disposition compared to using structured data alone. The authors noted that the model was

developed at a single hospital and that its performance in settings where nursing notes are brief, inconsistent, or recorded on paper had not been evaluated [11].

Wang et al. (2025) Prolonged wait times in the emergency department are a patient safety concern, but they are difficult to predict and prevent without early warning. Wang et al. developed interpretable machine learning models using ED data to predict which patients were likely to experience prolonged wait times, with a focus on making model outputs understandable to clinical and operational staff. They found that interpretable models could identify patients at risk of long delays while remaining transparent enough for staff to act on. The authors noted that the study was conducted at a single hospital and that the predictors of prolonged wait time may differ in other settings, including those with different staffing ratios or bed availability [12].

Song et al. (2025) Predicting how long a patient will stay in the emergency department and where they will ultimately go are important for operational planning, but existing tools often lack transparency. Song et al. developed interpretable machine learning models using ED data to predict length of stay and disposition decisions for adult patients, prioritising transparency in how the models reached their outputs. They found that interpretable models performed well on both outcomes while remaining explainable to clinical users, reinforcing that transparency is a practical requirement for clinical adoption rather than an optional feature. The authors noted that the models were developed at a single centre and that external validation in other ED settings, including lower-resource environments, had not been completed [13].

Teeple et al. (2023) Missing data is common in emergency department records, but its effect on the fairness of AI triage models is not well understood. Teeple et al. analysed emergency department data to examine how gaps in patient records affected the predictive performance of a machine learning triage model across different racial groups. They found that data missingness contributed to racial disparities in model performance, meaning that patients whose records were less complete were more likely to be inaccurately assessed by the AI tool. The authors noted that their findings came from a single institution and that the relationship between missingness and bias may vary across settings with different documentation practices and patient populations [14].

Huang et al. (2022) Racial bias in clinical machine learning models is a known concern, but it is unclear how consistently researchers identify and address it. Huang et al. conducted a scoping review of published clinical machine learning studies to assess how frequently racial bias was evaluated and whether mitigation methods were applied. They found that racial bias was commonly identified across clinical machine learning models but was inconsistently measured and rarely addressed in a systematic way. The authors noted that the lack of standardised methods for detecting and mitigating bias makes it difficult to compare findings across studies or to know whether any given model is safe to deploy in a new population [15].

Shortlist of 3 Candidate Workflow Constraints

1. Incomplete historical ED records [Technical Constraint]

Caribbean ED records may be incomplete, paper-based, or difficult to compile, which could limit the data available to train and test an AI tool designed to flag high-risk patients earlier.

2. Alert fatigue during busy shifts [Operational Constraint]

If the AI tool creates too many alerts or requires extra clicks, triage nurses may ignore it during crowded or high-pressure periods.

3. Limited bed availability [Structural Constraint]

An admission-likelihood flag can help staff plan earlier, but it cannot create more inpatient beds.

Stakeholders

1. Triage Nurse

Concern: Too many alerts or extra steps to use the system could slow triage.

Success: The tool provides a simple, timely risk flag that helps nurses identify high-risk patients earlier without slowing triage or replacing their final decision.

2. ED Consultant

Concern: The AI recommendation may be wrong, unclear, or may cause over-triage or under-triage.

Success: The consultant can use the AI flag as an early warning to review potentially high-risk patients sooner, while maintaining clinical oversight.

3. Porter / Support Staff

Concern: Earlier identification of high-risk patients may increase the need for rapid movement between triage, treatment areas and wards, or resuscitation spaces.

Success: Support staff get earlier notice so they can prepare stretchers, wheelchairs, and transport support when high-risk patients need to move quickly.

4. Admin Clerk

Concern: If registration or triage information is incomplete or paper-based, the model may not have enough reliable data and may affect model accuracy.

Success: Registration and triage information are recorded clearly enough for the tool to use existing data. Older records are cleaned and organized well enough for the model to learn patterns.

5. Patient Advocate

Concern: AI-assisted triage must be safe, fair, and should not reduce the role of human clinical judgement.

Success: Patients remain assessed by triage nurses, while the AI tool helps flag high-risk cases earlier so critical patients can be reviewed and monitored sooner.

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